Chapter 12

Medical Microbiology

Chapter Outline

12.1 Microbial Infections of the Human Body

12.2 Skin and Wound Infections

12.3 Respiratory Tract Infections

12.4 Gastrointestinal Tract Infections

12.5 Ocular Infections

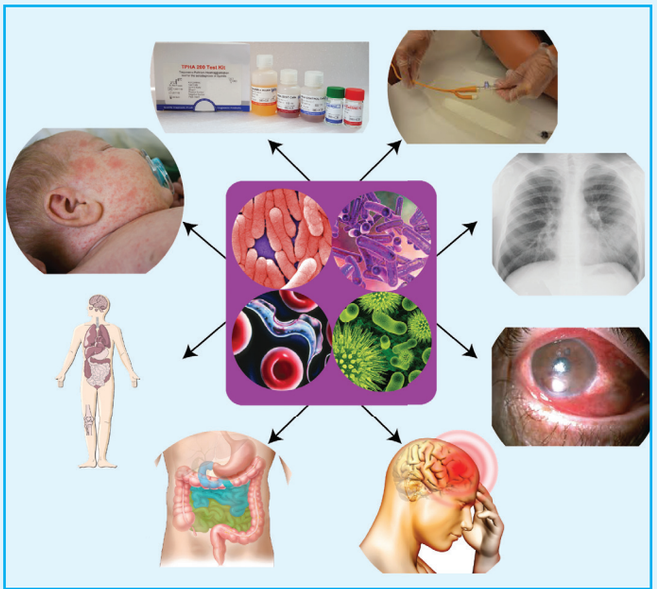
12.6 Urinary Tract Infections

12.7 Reproductive Tract Infections

12.8 Infections of the Nervous System

12.9 Systemic Infections

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Medical Microbiology or Clinical Microbiology plays an important role by providing the necessary diagnostic testing, means of epidemiological detection, and future innovation required in an era of emerging and reemerging infectious diseases.

Learning Objectives

After studying this chapter the student will be able,

- *To describe the importance of medical microbiology.*
- *To understand the types and sources of infections.*
- *To know the types of infectious diseases and virulence factors of the pathogen.*
- *To tell the etiological agents of skin wound respiratory, gastro intestinal, ocular, urinary, reproductive, nervous system and systemic infections.*
- *To know the causative agents of varoius human diseases and their portal of entry.*

12.1 Microbial Infections of the Human Body

Medical microbiology is the branch of microbiology which deals with prevention, diagnosis and treatment of infectious diseases. There are four kinds of microorganisms that cause infectious diseases. They are bacteria, fungi, parasites and viruses. Any disease that spreads from one host to another, either directly or indirectly is said to be a communicable disease. Chicken pox, measles, genital herpes, typhoid fever and tuberculosis are examples of such diseases, that are easily spread from one person to another.

A non communicable disease does not spread from one host to another. For example, *Clostridium tetani,* a soil